MIL-T-52501B(ME)
25 August 1981
SUPERSEDING
MIL-T-52501A
19 November 1975

MILITARY SPECIFICATION

TRAINING AID, SCALE, COORDINATE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a coordinate scale training aid panel with scale markings on two adjacent edges.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
LLL-B-810	- Building Board (Hardboard), Hard Pressed, Vegetable Fiber.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6910

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The training aid shall be a square panel, marked as shown on Figure 1.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Construction.

3.3.1 Building board. Building board shall conform to LLL-B-810, Type II, Surface 2, Finish C, Design a, size 1/8 inch thick.

3.3.1.1 Panel size. The panel size shall be 28.0 inches high x 28.0 inches wide x 0.125 inch thick.

3.3.2 Scale graduation marks. The graduation marks of the scales shall be 0.125 inch wide. Graduation marks representing 100-meter intervals shall be 2.0 inches high. Intermediate graduation marks representing 50-meter intervals shall be 1.125 inches high.

3.3.2.1 Scale intervals. The 1000-meter graduation mark shall be located flush against the panel edge as shown on Figure 1. The interval between all other graduation marks on the scale shall be 1.40 inches when measured from the center of one mark to the center of the next mark.

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3.3.2.2 Character sizes.

3.3.2.2.1 Scale numerals. Numerals on the scale representing 100-meter intervals shall be gothic sans serif 1.0 inch high and 0.187 inch wide.

3.3.2.2.2 Range finder ratio. The range finder ratio characters shall be gothic sans serif 1.50 inches high and 0.25 inch wide.

3.3.2.3 Tolerance. The tolerance on all dimensions shall be plus or minus 0.03.

3.4 Treatment and painting. The panel shall be treated and painted in accordance with MIL-T-704, Type A. The panel background shall be painted white and all markings shall be painted black.

3.5 Workmanship. Particular attention shall be given to neatness and legibility of all markings. Panels shall have smooth edges free from spots, blisters, tears, scuffs, and checks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspectio·.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The scales shall be examined as specified herein for the following major defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Marking not as specified.
104. Treatment and painting not as specified.
105. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall constitute one defect.
107. Scale bent.
108. Consolidation not as specified for Level A.
109. Gross weight or size exceeds the limitations of the box specification for Level A.
110. Strapping not as specified for Level A.
111. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each scale shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. Bending of the scale shall not be permitted.

5.1.1.1 Consolidation. The preserved scales shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c, style optional, in quantities not to exceed the weight limitation of the box specification.

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5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Scales, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified herein, shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Scales, preserved as specified in 5.1, shall be packed as specified for Level A except that the boxes shall conform to PPP-B-636, V3c, style optional. Quantities packed together shall not exceed the size or weight limitation of the box. Closure shall be in accordance with the box specification appendix.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The scale is a component of a map reading set and is used as an aid in teaching students the method of accurate location of position points within grid squares.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(c) When strapping shall be Finish A (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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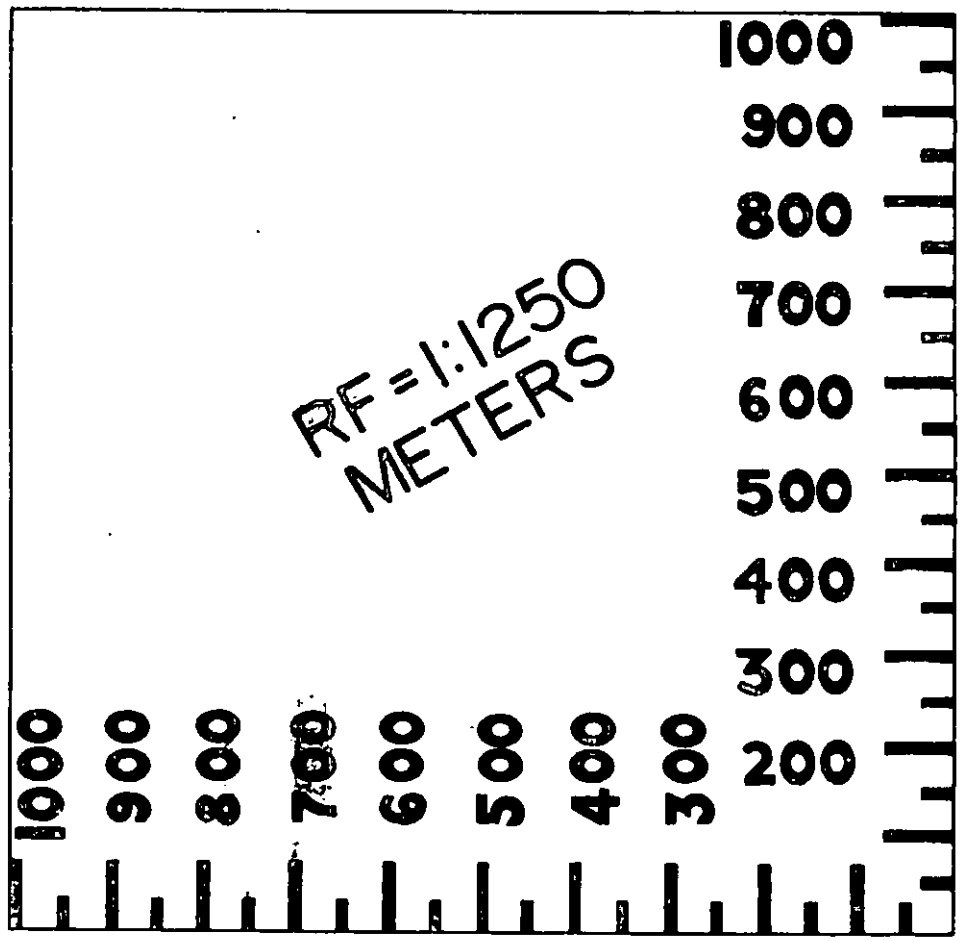


FIGURE 1. Training aid, scale coordinate.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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